Module 2: Cells, chemicals for life, transport and gas exchange

2.1 Cells and chemicals for life

This section provides learners with a knowledge and understanding of how the use of microscopy allowed the development of the cell theory, which is a unifying concept in biology. It also focuses on the importance of microscopy in investigating the structure of eukaryotic and prokaryotic cells. Learners will also gain an insight into some of the advanced microscopic techniques used by cell biologists today.

The structure of animal cells is studied in the context of the histology of mammalian blood as revealed by light microscopy and is contrasted with the structure of typical plant cells.

An understanding of the importance of enzyme-catalysed reactions is illustrated by the biochemical processes that prevent excessive blood loss, and learners investigate how first-aid and medical intervention can assist the body's natural mechanisms for preventing blood loss.

In addition to enzymes, this section covers the structures and functions of other biologically important molecules in animals and plants, including DNA, natural polymers, and the different mechanisms for transporting molecules into and out of living cells. Learners gain an appreciation of the importance of water in living organisms and as the essential medium in which transport and exchange takes place.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.